

Expressivity of Description Logics and the Corresponding OWL Profiles

From DL letters to GraphDB rulesets, ECMO, and openly available reasoners

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JRC — Reasoning notes

Languages

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(with an empirical check)
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How the naming works

A DL name is a concatenation of letters

A *base language* ($\mathcal{AL}$, $\mathcal{EL}$, $\mathcal{FL}$, or the shorthand $\mathcal{S}$) followed by one letter per additional constructor.

$$SROIQ^{(\mathcal{D})} = \underbrace{\mathcal{S}}_{\mathcal{ALC}+\text{trans.}} + \underbrace{\mathcal{R}}_{\text{role axioms}} + \underbrace{\mathcal{O}}_{\text{nominals}} + \underbrace{\mathcal{I}}_{\text{inverses}} + \underbrace{\mathcal{Q}}_{\text{qual. card.}} + \underbrace{(\mathcal{D})}_{\text{datatypes}}$$

Notation used throughout

A atomic concept (class name) C, D concept expressions R, S roles (properties) a, b individuals.

Turtle snippets assume the usual `rdf:`, `rdfs:`, `owl:`, `xsd:` prefixes and a local prefix `:`.

DL identifiers at a glance

Id	Construct	DL syntax	OWL 2 vocabulary
<i>Base languages</i>			
$\mathcal{AL}$	attributive language	$\top, \perp, \neg A, C \sqcap D, \forall R.C, \exists R.\top$	intersectionOf, allValuesFrom, ...
$\mathcal{EL}$	existential language	$\top, C \sqcap D, \exists R.C$	intersectionOf, someValuesFrom
$\mathcal{FL}_0$	frame language	$C \sqcap D, \forall R.C$	intersectionOf, allValuesFrom
$\mathcal{S}$	$\mathcal{ALC}$ + transitive roles	$\mathcal{ALC}, \text{Trans}(R)$	TransitiveProperty
<i>Constructor letters</i>			
$\mathcal{C}$	complex negation	$\neg C$	complementOf
$\mathcal{U}$	union	$C \sqcup D$	unionOf
$\mathcal{E}$	full existential	$\exists R.C$	someValuesFrom
$\mathcal{H}$	role hierarchy	$R \sqsubseteq S$	subPropertyOf
$\mathcal{R}$	complex role inclusions	$R_1 \circ R_2 \sqsubseteq S, \exists R.\text{Self}$	propertyChainAxiom, hasSelf
$\mathcal{O}$	nominals	$\{a, b\}$	oneOf, hasValue
$\mathcal{I}$	inverse roles	R^-	inverseOf
$\mathcal{N}$	unqualified cardinality	$\geq n R, \leq n R$	min/maxCardinality
$\mathcal{Q}$	qualified cardinality	$\geq n R.C, \leq n R.C$	min/maxQualifiedCardinality
$\mathcal{F}$	functional roles	$\top \sqsubseteq \leq 1 R$	FunctionalProperty
$(\mathcal{D})$	datatypes	$\exists U.d, \text{facets}$	DatatypeProperty, withRestrictions
<i>DL-Lite</i>	FOL-rewritable fragments	$B \sqsubseteq C, B ::= A \mid \exists R$	OWL 2 QL (<i>DL-Lite</i> _{$\mathcal{R}$})

Base languages and constructor letters

Base languages: $\mathcal{FL}_0$, $\mathcal{AL}$, $\mathcal{EL}$

- $\mathcal{FL}_0$ / $\mathcal{FL}^-$ — conjunction and value restriction only ($\mathcal{FL}^-$ adds $\exists R.T$). Historical; no OWL fragment.
- $\mathcal{AL}$ — the classical baseline: $\top$, $\perp$, *atomic* negation $\neg A$, $\sqcap$, $\forall R.C$, *unqualified* $\exists R.T$.
- $\mathcal{EL}$ — $\top$, $\sqcap$, *full* $\exists R.C$. No negation, no disjunction, no $\forall$. **Base of OWL 2 EL.**

$\mathcal{AL}$: $\text{Parent} \sqsubseteq \text{Person} \sqcap \neg \text{Robot} \sqcap \forall \text{hasChild}.\text{Person} \sqcap \exists \text{hasChild}.\top$

```
:Parent rdfs:subClassOf [ a owl:Class ;  
  owl:intersectionOf ( :Person  
    [ a owl:Class ; owl:complementOf :Robot ]  
    [ a owl:Restriction ; owl:onProperty :hasChild ;  
      owl:allValuesFrom :Person ]  
    [ a owl:Restriction ; owl:onProperty :hasChild ;  
      owl:someValuesFrom owl:Thing ]  
  ) ] .
```

$\mathcal{EL}$: $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild}.\text{Person}$

```
:Parent owl:equivalentClass [  
  a owl:Class ;  
  owl:intersectionOf ( :Person  
    [ a owl:Restriction ;  
      owl:onProperty :hasChild ;  
      owl:someValuesFrom :Person ] ) ] .
```

Boolean letters: $\mathcal{C}$, $\mathcal{U}$, $\mathcal{E}$ — and $\mathcal{S}$

- $\mathcal{C}$ full negation. $\mathcal{AL} + \mathcal{C} = \mathcal{ALC}$, closed under Booleans $\Rightarrow \mathcal{U}$ and $\mathcal{E}$ come for free (De Morgan).
- $\mathcal{S}$ is not a constructor: shorthand for $\mathcal{ALC}_{R^+}$ ($\mathcal{ALC} +$ transitive roles).

$\mathcal{C}$: Childless $\equiv \neg \exists \text{hasChild}.\top$

```
:Childless owl:equivalentClass [ a owl:Class ;  
  owl:complementOf [ a owl:Restriction ;  
    owl:onProperty :hasChild ;  
    owl:someValuesFrom owl:Thing ] ] .
```

$\mathcal{U}$: Person $\equiv \text{Man} \sqcup \text{Woman}$

```
:Person owl:equivalentClass [  
  a owl:Class ; owl:unionOf ( :Man :Woman ) ] .
```

$\mathcal{E}$: Researcher $\sqsubseteq \exists \text{worksFor}.\text{Institution}$

```
:Researcher rdfs:subClassOf [  
  a owl:Restriction ; owl:onProperty :worksFor  
    ;  
  owl:someValuesFrom :Institution ] .
```

$\mathcal{S}$: hasAncestor $\circ \text{hasAncestor} \sqsubseteq \text{hasAncestor}$

```
:hasAncestor a owl:TransitiveProperty .
```

Role letters: $\mathcal{H}$ and $\mathcal{R}$

- $\mathcal{H}$ role hierarchy: $\text{hasMother} \sqsubseteq \text{hasParent} \rightarrow \text{rdfs:subPropertyOf}$
- $\mathcal{R}$ (subsumes $\mathcal{H}$): role chains under a *regularity* condition, reflexive / irreflexive / disjoint roles, local reflexivity $\exists R.\text{Self}$

$\text{hasParent} \circ \text{hasParent} \sqsubseteq \text{hasGrandparent}$

$\text{Disj}(\text{hasParent}, \text{hasSpouse})$

$\text{Narcissist} \equiv \exists \text{loves}.\text{Self}$

```
:hasGrandparent owl:propertyChainAxiom
    ( :hasParent :hasParent ) .
:hasParent owl:propertyDisjointWith :hasSpouse .
:hasPart a owl:ReflexiveProperty .
:hasParent a owl:IrreflexiveProperty .

:Narcissist owl:equivalentClass [
    a owl:Restriction ; owl:onProperty :loves ;
    owl:hasSelf "true"^^xsd:boolean ] .
```

Individuals and inverses: $\mathcal{O}$ and $\mathcal{I}$

$\mathcal{O}$ — nominals

$\text{Continent} \equiv \{\text{Africa}, \text{Americas}, \dots, \text{Oceania}\}$

$\text{Italian} \equiv \exists \text{hasNationality}.\{\text{Italy}\}$

```
:Continent owl:equivalentClass [ a owl:Class ;  
  owl:oneOf ( :Africa :Americas :Antarctica  
              :Asia :Europe :Oceania ) ] .
```

```
:Italian owl:equivalentClass [  
  a owl:Restriction ;  
  owl:onProperty :hasNationality ;  
  owl:hasValue :Italy ] .
```

$\mathcal{I}$ — inverse roles

$\text{hasChild} \equiv \text{hasParent}^{-}$

$\text{Child} \sqsubseteq \exists \text{hasChild}^{-}.\text{Person}$

```
:hasChild owl:inverseOf :hasParent .
```

```
:Child rdfs:subClassOf [  
  a owl:Restriction ;  
  owl:onProperty [ owl:inverseOf :hasChild ] ;  
  owl:someValuesFrom :Person ] .
```

Counting and data: $\mathcal{N}$, $\mathcal{Q}$, $\mathcal{F}$, ($\mathcal{D}$)

$\mathcal{N}$ unqualified: $\text{Person} \sqsubseteq \leq 2 \text{ hasParent}$

```
:Person rdfs:subClassOf [ a owl:Restriction ;  
  owl:onProperty :hasParent ;  
  owl:maxCardinality "2"^^xsd:nonNegativeInteger ]  
.
```

$\mathcal{Q}$ qualified (OWL 2): $\text{Hand} \sqsubseteq \geq 5 \text{ hasPart.Finger}$

```
:Hand rdfs:subClassOf [ a owl:Restriction ;  
  owl:onProperty :hasPart ; owl:onClass :Finger ;  
  owl:minQualifiedCardinality "5"^^xsd:  
    nonNegativeInteger ] .
```

$\mathcal{F}$ functional: $\top \sqsubseteq \leq 1 R$ — weaker than $\mathcal{N}$, kept in *SHIF* (OWL Lite) for complexity

```
:hasBiologicalMother a owl:FunctionalProperty .  
:hasSSN a owl:InverseFunctionalProperty .
```

($\mathcal{D}$) datatypes & facets:

$\text{Adult} \equiv \text{Person} \sqcap \exists \text{hasAge}.(\geq 18)$

```
:Adult owl:equivalentClass [ a owl:Class ;  
  owl:intersectionOf ( :Person  
    [ a owl:Restriction ; owl:onProperty :  
      hasAge ;  
        owl:someValuesFrom [ a rdfs:Datatype ;  
          owl:onDatatype xsd:integer ;  
          owl:withRestrictions  
            ( [ xsd:minInclusive 18 ] ) ] ) ] ) ] .
```

Hierarchy of counting

$\mathcal{F} \subset \mathcal{N} \subset \mathcal{Q}$ ($\mathcal{N} = \mathcal{Q}$ with $C = \top$)

The *DL-Lite* family

DL-Lite: designed bottom-up

- Letter-based DLs: *top-down* — start rich, keep what stays decidable.
- *DL-Lite* (Calvanese et al., *JAR* 2007):
bottom-up, one criterion:

CQ answering must be FOL-rewritable

i.e. reducible to an SQL query over the data alone.

- Every restriction follows from that criterion.

Syntax

$B ::= A \mid \exists R$	basic concept
$R ::= P \mid P^-$	basic role
$C ::= B \mid \neg B$	general concept
$E ::= R \mid \neg R$	general role

$\exists R$ is *unqualified* ($\exists R.\top$).

The essential asymmetry

Only a *basic* concept on the left of $\sqsubseteq$; negation only on the right. The canonical model can be *folded into the query* instead of materialised over the data.

core — inclusions +

disjointness

$\text{Professor} \sqsubseteq \exists \text{teaches}$
 $\exists \text{teaches} \sqsubseteq \text{Professor}$ (domain)
 $\exists \text{teaches}^- \sqsubseteq \text{Course}$ (range)
 $\text{Professor} \sqsubseteq \neg \text{Student}$

```
:teaches rdfs:domain :Professor ;  
      rdfs:range :Course .  
:Professor owl:disjointWith :  
      Student .
```

$\mathcal{F}$ — functionality

(func hasSupervisor)
(func hasMatricula⁻)
No role inclusions.

```
:hasSupervisor  
  a owl:FunctionalProperty .  
:hasMatricula  
  a owl:  
      InverseFunctionalProperty  
  .
```

$\mathcal{R}$ — role inclusions

$\text{hasSupervisor} \sqsubseteq \text{knows}$
 $\text{hasSupervisor}^- \sqsubseteq \text{supervises}$
 $\text{hasSupervisor} \sqsubseteq \neg \text{hasStudent}$
No functionality. **Logic of OWL 2 QL.**

```
:hasSupervisor  
  rdfs:subPropertyOf :knows ;  
  owl:propertyDisjointWith :  
      hasStudent .
```

OWL 2 QL also admits $\exists R.A$ in superclass position — a conservative extension: $A \sqsubseteq \exists R.B \rightsquigarrow A \sqsubseteq \exists R_B$,
 $R_B \sqsubseteq R$, $\exists R_B^- \sqsubseteq B$.

$DL-Lite_{\mathcal{A}}$ and the Horn / Boolean extensions

$\mathcal{F} + \mathcal{R}$ **unrestricted** $\Rightarrow$ no FOL-rewritability, ExpTime-hard (Artale et al., *JAIR* 2009). $DL-Lite_{\mathcal{A}}$ restores it: *a functional role may never be specialised*.

$\Rightarrow$ OWL 2 QL keeps role inclusions and drops `Functional/InverseFunctionalProperty` altogether.
($DL-Lite_{\mathcal{A}}$ also splits objects from *attributes* — ancestor of object vs. datatype properties.)

Fragment	Satisfiability	CQ data compl.	Left-hand side of $\sqsubseteq$
$DL-Lite_{core}$, $DL-Lite_{\mathcal{F}}$, $DL-Lite_{\mathcal{R}}$	NLogSpace-c.	AC^0	a single basic concept B
$DL-Lite_{krom}$	NLogSpace-c.	AC^0	binary clauses
$DL-Lite_{horn}$	PTime-c.	PTime	$B_1 \sqcap \dots \sqcap B_n$
$DL-Lite_{bool}$	NP-c.	coNP	arbitrary Boolean combination

The $AC^0 \rightarrow$ PTime jump at the Horn line is where the TBox can no longer be compiled into the query and the reasoner must touch the data.

Why FOL-rewritability is the point

Given a CQ q and a TBox $\mathcal{T}$, PerfectRef produces a UCQ $q_{\mathcal{T}}$ such that for every ABox $\mathcal{A}$:

$$\text{cert}(q, \langle \mathcal{T}, \mathcal{A} \rangle) = \text{ans}(q_{\mathcal{T}}, \mathcal{A})$$



- Ontology compiled into the *query*, never into the data $\Rightarrow$ OBDA (Mastro, Ontop); the reason OWL 2 QL exists.
- Cost: the rewriting may grow exponentially in the size of the *query* (not of the data) $\Rightarrow$ heavy optimisation and pruning in practice.

Named DLs and OWL

Named DLs and their OWL counterparts

Description Logic	OWL language	Complexity (concept sat.)
$\mathcal{EL} / \mathcal{EL}^{++}$	OWL 2 EL	PTime-complete
$DL\text{-}Lite_{\mathcal{R}}$	OWL 2 QL	NLogSpace; AC^0 data compl. for CQs
DLP / pD^* -style	OWL 2 RL	PTime-complete
ALC	—	ExpTime-complete
$SHIF^{(\mathcal{D})}$	OWL 1 Lite	ExpTime-complete
$SHOIN^{(\mathcal{D})}$	OWL 1 DL	NExpTime-complete
$SROIQ^{(\mathcal{D})}$	OWL 2 DL	N2ExpTime-complete
unrestricted, RDF semantics	OWL 2 Full	undecidable

OWL Lite was not “lite”

It is $SHIF^{(\mathcal{D})}$: same worst-case complexity as $SHIF$, and most of OWL DL can be encoded into it — hence effectively abandoned in OWL 2.

The three OWL 2 profiles

Three profiles, three targets — pairwise incomparable

OWL 2 EL ($\mathcal{EL}^{++}$)

Target: very large terminologies (SNOMED CT, GO, NCIt).

Keeps: $\sqcap$, $\exists R.C$, nominals, chains, Self, domain/range, keys, datatypes.

Drops: $\forall$, $\sqcup$, $\neg$, inverses, cardinality, functional properties.

Classification is **polynomial**.

OWL 2 QL ($DL-Lite_{\mathcal{R}}$)

Target: query answering over RDBMS via rewriting (OBDA, Ontop).

Left: class or $\exists R.T$; **right:** class, $\exists R.C$, $\neg$.

Drops: $\sqcup$, chains, functional & transitive props, nominals, keys.

AC⁰ data complexity.

OWL 2 RL

Target: rule engines, forward chaining over RDF (Datalog).

Left: $\sqcap$, $\sqcup$, $\exists R.C$, nominals;
right: $\sqcap$, $\forall R.C$, ≤ 1 , ≤ 0 , $\neg$.

No existential on the right $\Rightarrow$ no blank-node generation.

PTime; degrades gracefully on OWL Full graphs.

Profile comparison

Construct	EL	QL	RL
$C \sqcap D$	✓	partial	✓
$C \sqcup D$	×	×	subclass pos. only
$\neg C$	×	superclass pos. only	superclass pos. only
$\exists R.C$	✓	superclass pos. only	subclass pos. only
$\forall R.C$	×	×	superclass pos. only
Nominals $\{a\}$	✓	×	✓
Inverse roles R^-	×	✓	✓
Cardinality restrictions	×	×	$\leq 1, \leq 0$ only
Functional properties	×	×	✓
Transitive properties	✓	×	✓
Property chains	✓	×	✓
$\exists R.\text{Self}$	✓	×	×
Keys	✓	×	✓
Data complexity	PTime	AC^0	PTime

Rule-based entailment: the GraphDB rulesets

Triple stores reason with rules, not DL reasoners

GraphDB rulesets: empty, rdfs, rdfsplus, owl-horst, owl-max, owl2-ql, owl2-rl (+ -optimized variants).

The names suggest a mapping to the languages above — **but it is loose**:

1. **Rule engine.** Horn rules (.pie), forward chaining, closure fully materialised at load time.
2. **RDF-based semantics.** Rules match triples, not axioms; classes may be instances; no Direct Semantics.
3. **No syntactic check.** Any graph is accepted, in-profile or not.

Consequence

No ruleset implements a full DL. Each is a *Horn fragment* of some DL: sound, but incomplete w.r.t. the logic whose vocabulary it accepts.

Mapping rulesets to DL families

Ruleset	Closest DL / profile	DL axioms covered
rdfs	$\rho df \subset DL-Lite_{\mathcal{R}}$	$A \sqsubseteq B, P \sqsubseteq Q, \exists P \sqsubseteq A$ (domain), $\exists P^- \sqsubseteq A$ (range)
rdfsplus	Horn- $\mathcal{SHI}$ ($\approx$ DLP)	+ $P \equiv Q^-, P \equiv P^-$ (symmetry), $\text{Trans}(P)$
owl-horst	$\rho D^* \approx \text{Horn-}\mathcal{SHOIF}$	+ $a \approx b, \text{Func}(P), \text{Func}(P^-), \exists P.C \sqsubseteq A, A \sqsubseteq \forall P.C, A \equiv \exists P.\{o\}, A \equiv B$
owl-max	Horn- $\mathcal{SHIF}^{(\mathcal{D})}$ (OWL Lite syntax)	+ $B \sqcup C \sqsubseteq A$, intersections, $\leq 1 P$
owl2-rl	OWL 2 RL $\subset \text{Horn-}\mathcal{SROIQ}^{(\mathcal{D})}$	+ property chains, keys, disjoint properties, qualified ≤ 1
owl2-ql	OWL 2 QL $\approx DL-Lite_{\mathcal{R}}$	$A \sqsubseteq B, A \sqsubseteq \exists P.B$, role inclusions, inverses, disjointness

Reference points: $\mathcal{SHIF}^{(\mathcal{D})} = \text{OWL Lite}$, $\mathcal{SHOIN}^{(\mathcal{D})} = \text{OWL 1 DL}$, $\mathcal{SROIQ}^{(\mathcal{D})} = \text{OWL 2 DL}$. **No ruleset for OWL 2 EL.**

Why every ruleset is a Horn fragment

No disjunctive consequences

A rule derives $A(a)$, never $A(a) \vee B(a)$.

$\text{Staff} \equiv \text{Academic} \sqcup \text{TechnicalStaff}$

only $\text{Academic} \sqsubseteq \text{Staff}$ is usable; the converse (tableau case analysis) is lost.

No new individuals

Rules never invent a witness.

$A \sqsubseteq \exists P.B$ is accepted and stored,
but yields no P -successor — nor anything that
would follow from one.

Together these two limitations account for most of the distance from the full logics.

Ruleset by ruleset

RDFS Not a DL (metamodelling). DL-part $\subset DL-Lite_{\mathcal{R}} \cap \mathcal{EL}$.

RDFS-Plus Role axioms of *SHI* (inverse, transitive) on top of the hierarchy; no new concept constructors. Within DLP (Grosz et al., WWW 2003).

OWL-Horst Ter Horst's pD^* (JWS 2005): “if”-semantics over RDFS. $\approx$ Horn-*SHOIF*; sound, incomplete for *SHOIN*. GraphDB goes beyond textbook pD^* on `intersectionOf`.

OWL-Max OWL Lite syntax, Horn consequences only. Adds subclass-side `unionOf`, `$\leq 1 \rightarrow$  sameAs`, more consistency checks. Exact diff: compare the .pie files.

OWL 2 RL Derived from DLP and pD^* . Sound & complete for ground atoms *on in-profile ontologies* (Thm. PR1); sound but incomplete outside.

OWL 2 QL GraphDB *materialises* instead of rewriting — forgoes the profile's main benefit; more permissive than the profile.

-optimized: drops axiomatic triples and `rdfs:Resource` conclusions (+ a few more). The documented list does *not* include domain/range.

Empirical check: set-up

- Small university KG, one construct per axiom block.
- Five repositories: rdfsplus, owl-horst, owl-max, owl2-ql, owl2-rl (all -optimized, owl:sameAs on).
- Two runs; Horst replaced by RL in the second; all else identical.

The key trick: an individual with no asserted type

```
ex:teaches a owl:ObjectProperty ;  
           rdfs:domain ex:Academic ; rdfs:range ex:Course .  
ex:alice   ex:teaches ex:kg101 .      # no rdf:type for ex:alice
```

Every class membership of `ex:alice` must be *inferred*.

Empirical check: results

	Construct tested	RDFS+	Horst ^e	Max	QL	RL
Q1	domain, subclass	✗ ^a	✓	✓	✓	✓
Q2	subproperty, range	✗ ^a	✓	✓	✓	✓
Q3	inverse, symmetric	✓	✓	✓	✓	✓
Q3	transitive	✓	✓	✓	✗	✓
Q4	equivalent class / property	— ^b	✓	✓	✓	✓
Q5	sameAs via (inverse-)functional prop.	✗	✓	✓	✗	✓
Q6	hasValue, some/allValuesFrom	✗	✓	✓	✗	✓ ^f
Q7	intersection, subclass side	✗	✓ ^c	✓	✓ ^d	✓
Q7	intersection, superclass side	✗	✓	✓	✓	✓
Q7	union, subclass side	✗	✗	✓	✗	✓
Q8	property chain	✗	✗	✗	✗	✓
Q9	keys	✗	✗	✗	✗	✓
Q10	implicit triples	193	337	361	1596	1263

^a unexpected. ^b not observable (depends on domain). ^c beyond textbook PD^* . ^d beyond OWL 2 QL profile. ^e first run only. ^f includes a classification the RL profile does not license.

On this test bed: owl2-rl $\supset$ owl-max $\supset$ owl-horst — though not nested by design. **RL is the only regime that passes every test.**

Five deviations from expectations

1. **rdfsplus-optimized applies neither domain nor range.** Inverse/symmetric/transitive work, so the ruleset is active; same symptom seen on RDFS. Not explained by the docs; owl2-rl-optimized is fine. *To confirm on the non-optimised ruleset.*
2. **OWL-Horst handles intersection in both directions** — textbook pD^* does not.
3. **OWL 2 QL derives membership in an intersection** — a subclass-side use $DL-Lite_{\mathcal{R}}$ does not allow.
4. **OWL 2 RL classifies beyond its profile.** Supervisor $\equiv \exists \text{supervises. PhDStudent}$ is out of profile, yet `ex:alice` is typed as a supervisor. *The profile restricts what is guaranteed, not what fires.*
5. **OWL 2 rulesets materialise far more** (QL 1596, RL 1263 vs. 193–361): both type everything as `owl:Thing`; QL > RL despite being weaker.

- **Names are not guarantees.** Weaker than the name (RDFS-Plus) or stronger than the profile (Horst, QL). **Test the constructs you rely on.**
- **DL-illegal input is silently accepted.** `InverseFunctionalProperty` on a datatype property is outside *SHOIN/SROIQ*, yet yields `owl:sameAs`.
- **The gap to a DL reasoner is measurable.** Compare inferred class assertions with HermiT / Konclude (or ELK for the EL part).
- **Materialisation cost is not monotone in expressivity.** $QL > RL \gg Max$.
- **Beyond RDFS-Plus, OWL 2 RL is the safe default** — only regime passing every test, only one with chains and keys.

Case study: ECMO 0.3.0

ECMO 0.3.0 against the GraphDB rulesets

- JRC ontology network for **emergency and crisis management**.
- Built on the **Sendai Framework** terminology; aligned to **DOLCE+DnS Ultralite** (DUL) through d0.
- Assessed modules: hub, core, core pattern, foundation, disaster, risk.
- Other domain modules and the inferred-axioms file not inspected $\Rightarrow$ *indicative, not a complete profile*.

The defined class carrying most semantic weight

$\text{Disaster} \equiv \text{HazardousEvent} \sqcap \exists \text{hasImpact} . (\text{Impact} \sqcap \exists \text{hasImpactIntensity} . \{\text{VeryHighImpactIntensity}\})$

Only the right-to-left ($\sqsupseteq$) half is rule-expressible — and only OWL 2 RL guarantees it. It lets a store classify a hazardous event as a disaster from its reported impacts.

ECMO constructs and ruleset support

Construct	Example in ECMO	RDFS	RDFS+	Horst	Max	QL	RL
RDFS core	subclass, subproperty, domain, range	✓	✓	✓	✓	✓	✓
Inverse properties	dozens of pairs, e.g. hasHazardDeterminant		✓	✓	✓	✓	✓
Transitive properties	subsumedUnder, specializes, includesEvent		✓	✓	✓	✗	✓
Symmetric property	isComplementaryTo		✓	✓	✓	✓	✓
hasValue, superclass side	DirectlyAffectedPopulation $\sqsubseteq$ $\exists$ hasAffectednessType.{Direct}			✓	✓	✗	✓
owl:sameAs	the HIPS hazard individuals			✓	✓	✗	✓
Defined class	Disaster ($\sqcap$, nested $\exists$, hasValue)			partial	partial	✗	✓ ^a
Union, subclass side	Eventuality $\equiv$ Event $\sqcup$ EventType $\sqcup$ Situation			✗	✓	✗	✓
Union, superclass side	Assessment $\sqsubseteq$ Eventuality $\sqcup$ InformationEntity	✗	✗	✗	✗	✗	✗
$\exists$, superclass side	Risk $\sqsubseteq$ $\exists$ hasHazardDeterminant.Hazard		no witness generated				
Qualified cardinality	Assessment $\sqsubseteq$ ≥ 1 assesses.Entity	✗	✗	✗	✗	✗	✗
Property chains	isSettingFor $\circ$ isAbout $\sqsubseteq$ assesses					✗	✓
Disjointness	Frequent/InfrequentDisaster; DUL: Description $\sqcap$ Situation $\sqsubseteq$ $\perp$		checks only (Max, RL)				
Punning	hazard, risk and disaster subtypes		transparent to rules				

^a classifying direction ($\sqsupseteq$) only.

ECMO's DL expressivity — and one out-of-DL axiom

Nominals, inverse & transitive roles, chains, qualified cardinality, superclass union, datatypes $\Rightarrow$ $SROIQ^{(D)} = \text{OWL 2 DL}$, outside all three profiles.

Strictly, one axiom leaves OWL 2 DL

foundation:assesses is defined by a property chain $\Rightarrow$ *non-simple*; OWL 2 DL forbids cardinality restrictions on non-simple roles.

```
foundation:Assessment rdfs:subClassOf [ a owl:Restriction ;  
    owl:minQualifiedCardinality "1"^^xsd:nonNegativeInteger ;  
    owl:onClass dul:Entity ; owl:onProperty foundation:assesses ] .
```

Fix

$\geq 1 R.C \equiv \exists R.C$, and existentials *are* allowed on non-simple roles: replace with `owl:someValuesFrom` `dul:Entity` — same meaning, OWL 2 DL restored.

What each ruleset delivers for ECMO

- RDFS-Plus** *Minimum useful regime:* DnS hierarchy via transitive `subsumedUnder/specializes`, named inverses, domain/range. Verify domain/range on the deployed ruleset (see anomaly).
- OWL-Horst** + `hasValue` classifications (affected population, risk profile); merges HIPS hazards via `sameAs`.
- OWL-Max** + union: every event/situation a `d0:Eventuality` (largely redundant with domain axioms).
- OWL 2 RL** **Only one** to fire the chains to DUL setting/aboutness, reliably classify `Disaster`, and check disjointness guards.
- OWL 2 QL** **Unsuitable:** no transitivity $\Rightarrow$ whole `subsumedUnder` hierarchy lost; no chains, no `hasValue`.
- None** Superclass union, superclass existentials (four determinants of `Risk`, `Signal`, `EventValidation`), cardinality $\Rightarrow$ need a DL reasoner.

- **Imports are not followed.** Load all modules and DUL/d0 explicitly — without DUL, chains have nothing to fire on and disjointness guards are absent.
- **Import closure incomplete as distributed.** 0.3.0-alignments/ missing; frames (imported by `ecmo-core`) has no catalog entry.
- **Recommended:** OWL 2 RL in the store + offline DL reasoner (Konclude / HermiT) for non-Horn consequences. `ecmo-inferredaxioms.ttl` may already be that output (not verified).
- **Modelling point:** `risk:Risk  $\sqsubseteq$  hazard:Hazard` $\Rightarrow$ every risk typed as a hazard — at odds with risk as $f(\text{hazard, exposure, vulnerability, capacity})$.

Full profile with ROBOT (once the catalog is fixed)

```
robot merge --catalog catalog-v001.xml --input ecmo-core.ttl --output merged.owl
robot validate-profile --profile RL --input merged.owl --output rl-violations.txt
robot validate-profile --profile DL --input merged.owl --output dl-violations.txt
```

Openly available reasoners

Three warnings before the tables

1. **“Supports $\mathcal{X}$ ” is a claim about the calculus, not the code.** Gaps usually in datatypes, nominals, exotic role axioms.
2. **Maintenance, not expressivity, is the binding constraint.** A 2023 review: fewer than half of reasoners active in the previous three years; HermiT among the effectively abandoned (Abicht, arXiv:2309.06888).
3. **The comparative evidence is dated.** Last OWL Reasoner Evaluation: ORE 2015 (Parsia et al., *JAR* 2017). No common benchmark since.

OWL 2 DL: the *SROIQ* tier

Reasoner	Expressivity	Method	Notes
Konclude	$SROIQV^{(\mathcal{D})}$	tableau + saturation	C++, LGPLv3. $\mathcal{V}$ = nominal schemas. Won most DL tracks of ORE 2015
HermiT	$SROIQ^{(\mathcal{D})}$	hypertableau	Java. Upstream dead since 1.3.8; OWL API fork unsupported. Ships with Protégé
FaCT++	$SROIQ^{(\mathcal{D})}$	optimised tableau	C++. Last commit 2017; ships with Protégé
JFact	$SROIQ^{(\mathcal{D})}$	tableau	Java port of FaCT++, under OWL API umbrella
Openllet	$SROIQ^{(\mathcal{D})}$	tableau	Community fork of Pellet; sporadic updates
Pellet	$SROIQ^{(\mathcal{D})}$	tableau	Abandoned; use Openllet
Sequoia	$SRIQ$	consequence-based	Scala. No nominals — traded for scalability
Racer	$SRIQ^{(\mathcal{D})}$	tableau	Common Lisp, BSD-3. Abandoned
Vampire	FOL	superposition	General prover; OWL DL by translation

Realistic choices: **Konclude** (fastest, actively developed) or **HermiT** (OWL API-native, ubiquitous, unmaintained).

OWL 2 EL and OWL 2 QL tiers

EL tier			
ELK	$\mathcal{EL}^{++}$, safe nominals	concurrent cons.-based	Java, Apache 2.0. Partial $\neg$, $\sqcup$, datatypes; incremental. De-facto biomedical standard; warns on unsupported constructs
jcel	$\mathcal{EL}^{+}$	completion rules	Java. Protégé plugin
CEL	$\mathcal{EL}^{+}$	completion rules	Original large-scale EL reasoner; historical
ElePHant	$\mathcal{EL}^{+}$	cons.-based	C. Small memory/CPU, embedded use
Whelk	$\mathcal{EL} + \text{RL}$	cons.-based	Scala. EL TBox + RL ABox; in ROBOT / OBO toolchain
Snorocket	EL subset	completion rules	Java. Concrete domains; SNOMED CT tooling
QL tier (rewriting / OBDA)			
Ontop	$DL\text{-}Lite_{\mathcal{R}} + \text{RDFS}$	rewriting to SQL	Java, Apache 2.0, active (5.5.0, Feb. 2026). SPARQL over VKGs via R2RML
Mastro	$DL\text{-}Lite_{\mathcal{A}}$	rewriting to SQL	Sapienza. Academic licence, not open source
Clipper	Horn- $SHIQ$	to Datalog	Abandoned; DLV back end no longer distributed
Requiem	$\mathcal{ELHIO}$	resolution	Prototype; not maintained

QL is effectively single-vendor: **Ontop** is the only open, maintained rewriting system.

OWL 2 RL: rules and materialisation

Reasoner	Expressivity	Method	Notes
RDFox	OWL 2 RL, Datalog, SWRL	parallel materialisation	C++. Fastest in class; commercial, free academic licences
VLog	Datalog + existential rules	chase	Open source; OWL RL via axiom-to-rule translation
OWL-RL	OWL 2 RL	forward chaining	Python on RDFLib. Convenient, not fast
reasonable	OWL 2 RL subset	forward chaining	Rust + Python bindings; much faster than OWL-RL
Arachne	OWL RL	Rete	Scala. Built for GO causal activity models
Apache Jena	RDFS, partial OWL	fwd/bwd rules	Deliberately incomplete; ubiquitous
GraphDB	RDFS → OWL 2 RL/QL	forward chaining	Commercial + free edition. Horn fragments; names ≠ guarantees
EYE	N3 rules	Euler-path chaining	Prolog/Wasm, very active; OWL-adjacent rules

Hybrid / pay-as-you-go

PAGOdA: RDFox for the RL bulk, HermiT for the hard residue. **MORe**, **Chainsaw**: modular routing
ELK/HermiT (unmaintained). **TrOWL**: approximate DL into QL/EL.

Choosing a reasoner

Expressivity narrows the field; the remaining criteria usually decide it.

Scenario	Choice
Large terminology, EL-shaped	ELK — unmatched at SNOMED scale; incremental mode
Full OWL 2 DL, performance matters	Konclude ; pure Java/OWL API → HermiT (unmaintained)
Data in an RDBMS, ontology as query interface	Ontop
Large RDF graphs, forward chaining OK	RDFox if licence OK; else VLog, Arachne, or the store's own
Embedded / constrained hardware	ElepHant (EL) or LiRoT (RL subset)

Take-aways

1. DL names are **compositional**; *DL-Lite* is the exception — designed bottom-up for **FOL-rewritability**.
2. The OWL 2 profiles are **pairwise incomparable**: EL for big TBoxes, QL for OBDA, RL for rules over RDF.
3. Triple-store rulesets are **Horn fragments** under RDF semantics: no disjunction, no witnesses, no syntactic check.
4. **Names are not guarantees** — RDFS-Plus under-delivers, Horst/QL/RL over-deliver. Test what you rely on.
5. For **ECMO**: OWL 2 RL in the store + an offline DL reasoner; fix the cardinality-on-non-simple-role axiom and the import catalog.
6. Among reasoners, **maintenance** decides: Konclude, ELK, Ontop are the living references of their tiers.